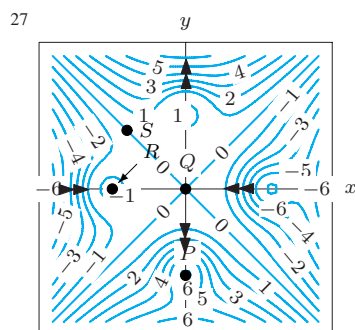


Saddle pts: $(1, 0), (-1, 4)$
Local min: $(1, 4)$

- 19 Saddle point: $(0, 0)$
Local max: $(1, 1), (-1, -1)$
21 $A = -2, B = 21$

- 23 (a) (a, b)
(b) $a = -1, b = 5$
(c) Local maximum

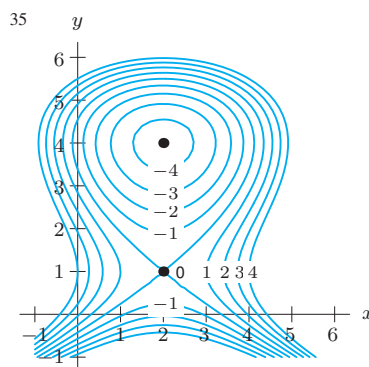
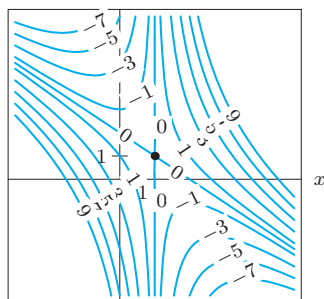
- 25 (a) Local maximum
(b) Saddle point
(c) Local minimum
(d) None of these



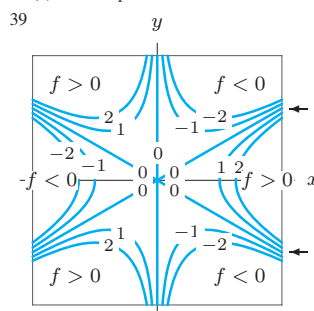
- 29 $(k\pi, l\pi)$,
for $k = 0, \pm 1, \pm 2, \dots$,
 $l = 0, \pm 1, \pm 2, \dots$
 $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$,
for $k = 0, \pm 1, \pm 2, \dots$,
 $l = 0, \pm 1, \pm 2, \dots$
 $(k\pi, l\pi), k = 0, \pm 1, \pm 2, \dots$,
 $l = 0, \pm 1, \pm 2, \dots$ are saddle points
If k and l are both even or k and l are both odd,
then $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$ are local max
If k is even but l is odd or k is odd but l is even,
then $(k\pi + \frac{\pi}{2}, l\pi + \frac{\pi}{2})$ are local min

- 31 $y = 0, \pm 2\pi, \pm 4\pi, \dots$ Local minima

- 33 (a) (a, b) is a saddle point.
(b)



- 37 (a) $(0, 0)$
(b) $D = -24x^2$
(c) Saddle point

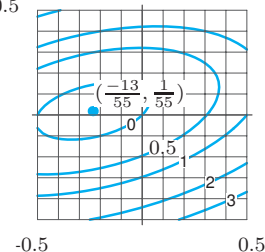


- 41 $(1, 3)$ could be saddle point
43 Can be saddle if f_{xy} large
45 $f(x, y) = 4 - (x - 2)^2 - (y + 3)^2$
47 False
49 True
51 False
53 True
55 False

Section 15.2

- 1 Mississippi:
87 – 88 (max), 83 – 87 (min)
Alabama:
88 – 89 (max), 83 – 87 (min)
Pennsylvania:
89 – 90 (max), 80 (min)
New York:
81 – 84 (max), 74 – 76 (min)
California:
100 – 101 (max), 65 – 68 (min)
Arizona:
102 – 107 (max), 85 – 87 (min)
Massachusetts:
81 – 84 (max), 70 (min)
3 Max: 30.5 at $(0, 0)$
Min: 20.5 at $(2.5, 5)$
5 Min = 0 at $(0, 0)$
(not on boundary)
Max = 2 at $(1, 1), (1, -1),$
 $(-1, -1)$ and $(-1, 1)$
(on boundary)
7 max = 1 at $(1, 0)$ and $(-1, 0)$
(on boundary)
min = -1 at $(0, 1), (0, -1)$
(on boundary)

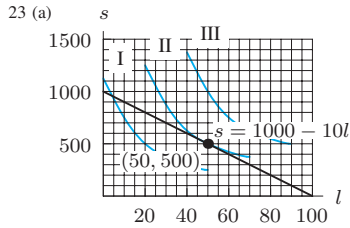
- 9 Global min
11 Global max = 0
No global min
13 (a) Local min: $(-13/55, 1/55)$
(b) 0.5



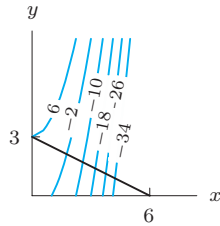
- 15 All edges $(32)^{1/3}$ cm
17 $l = w = h = 45$ cm
23 $q_1 = 300, q_2 = 225$.
25 (a) $L = \left[pA \left(\frac{a}{k} \right)^a \left(\frac{1}{b} \right)^{(a-1)} \right]^{1/(1-a-b)}$
 $K = \frac{l_a}{k_b} L$
(b) No
27 $y = 24x^2/49 - 2/7$
31 (a) $p = \sqrt{P_0 P_F}$
(b) $p_1 = \sqrt[3]{P_0^2 P_F}, p_2 = \sqrt[3]{P_0 P_F^2}$
33 Must have global maximum if continuous and R closed, bounded
35 Local max not necessarily global max
37 $f(x, y) = x^2 + y^2, R$ is $0 \leq x, y \leq 1$
39 False
41 False
43 False
45 True

Section 15.3

- 1 Min = $-\sqrt{2}$, max = $\sqrt{2}$
3 Max: 20 at $(-1, 2)$;
Min: 0 at $(1, -2)$
5 Min = -22, max = 22
7 Global min: $1/2$
No global max
9 Min = $-\sqrt{35}$, max = $\sqrt{35}$
11 Max: 4 at $(2, 2, 1), (2, -2, -1),$
 $(-2, 2, -1), (-2, -2, 1)$;
Min: -4 at $(-2, -2, -1), (-2, 2, 1),$
 $(2, -2, 1), (2, 2, -1)$
13 Max = $f(\frac{1}{\sqrt{5}}, \frac{3}{\sqrt{5}}) = 2\sqrt{5}$
Min = $f(-\frac{1}{\sqrt{5}}, -\frac{3}{\sqrt{5}}) = -2\sqrt{5}$
15 No global extrema
17 Max: $30 + \sqrt{10}$ at $(0, -\sqrt{10})$;
Min: -5 at $(\pm 3, -1)$
19 1 at $(1/2, 1/2)$
21 0.5



- (b) $s = 1000 - 10l$
- 25 (a) Min; max at endpt of constraint
 λ neg
(b) Max; min at endpt of constraint
 λ pos
- 27 $\Delta c/4; -\Delta c/4$
- 31 (a) $C = \$4349$
(b) \$182
- 33 (a) $W = 225$
 $K = 37.5$
(c) $W = 225$
 $K = 37.5$
 $\lambda = 0.29$
- 35 (a) No
(b) Yes
(c) $a + b = 1$
- 37 $x_1 = ((v_1)^{1/2} + (v_2)^{1/2})/(m(v_1)^{1/2})$
 $x_2 = ((v_1)^{1/2} + (v_2)^{1/2})/(m(v_2)^{1/2})$
- 39 (a) $f_1 = \frac{k_1}{k_1 + k_2}mg$, $f_2 = \frac{k_2}{k_1 + k_2}mg$
(b) Distance the mass stretches the top spring and compresses the lower spring
- 43 (a) Cost of producing quantity u when prices are p, q
(b) $2\sqrt{pqu}$
- 45 (a) $-5\lambda^2 + 15\lambda$
(b) 1.5, 11.25
(c) 11.25, 1.5
(d) same
- 47 (a) $S = \ln(a^a(1-a)^{(1-a)}) + \ln b - a \ln p_1 - (1-a) \ln p_2$
(b) $b = e^c p_1^a p_2^{(1-a)} / (a^a(1-a)^{(1-a)})$
- 49 Maximum value is 1
- 51 $f(x, y) = 3x + 4y$
- 53 $f(x, y) = x^2 + y^2$
- 55

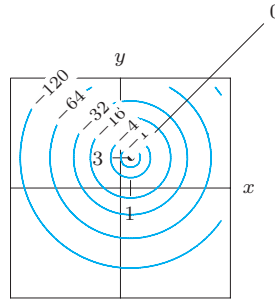


- 59 False
61 False
63 False
65 False
67 True
69 True

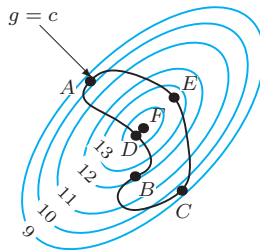
Chapter 15 Review

- 1 (3, -1), Saddle point

- 3 Saddle pt: (0, -5)
Local min: (2, -5)
- 5 (2, 1); local min
- 7 (2, 3): Local and global max
- 9 Local minimum: (1, 2)
Local maximum: (-1, -2)
Saddle points: (1, -2) and (-1, 2)
No global maximum or minimum
- 11 Minimum $f(-3/\sqrt{5}, 4/\sqrt{5}) = -5\sqrt{5}$,
Maximum $f(3/\sqrt{5}, -4/\sqrt{5}) = 5\sqrt{5}$
- 13 Min = 11.25; no max
- 15 Minimum $f(27.907, 23.256) = 1860.484$;
No maximum
- 17 Maximum = $f(48, 52) = 37,600$
No minimum
- 19 Max = 21, no min
- 21 No global extrema.
- 23 Maximum
- 25 Both
- 27 0.204
- 29 $\Delta c/2$; $\Delta c/(2\sqrt{2})$
- 31



- 33 (a) (i) Power function
(ii) Linear function
(b) $\ln N = 1.20 + 0.32 \ln A$
Agrees with biological rule
- 35 (a) $2c^2/3$
(b) $4c/3$
(c) $\lambda = m'(c)$
- 37 $c \left(\frac{aK}{(a+b)P_1} \right)^a \left(\frac{bK}{(a+b)P_2} \right)^b$
- 39 (a) Points A, B, C, D, E
(b) Point F
(c) Point D



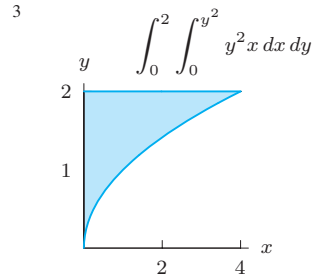
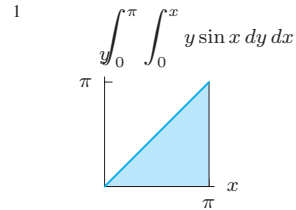
- 41 $f(x, y, z) = \frac{|Aa + Bb + Cc + D|}{\sqrt{A^2 + B^2 + C^2}}$
- 43 $K = 20$
 $L = 30$
 $C = \$7,000$

- 47 Base: 4 cm by 4 cm
Height: 2 cm
- 49 $-1/2$
- 51 (a) $x = 1/4 - a$, $y = 1$, saddle point
- 53 (a) 21.0208
(b) $g = 20.5, 21.2211$; $g = 20.2, 21.1008$
(c) $g = 20.5, 21.2198$; $g = 20.2, 21.1007$

Section 16.1

- 1 24; 43.5
- 3 Over: Approx 137
Under: Approx 60
- 5 about 2300
- 7 Positive
- 9 Zero
- 11 Zero
- 13 Positive
- 15 25.2°C
- 17 Need f nonnegative everywhere
- 19 $f(x, y) = 5 - x - y$; R is square with vertices $(\pm 1, \pm 1)$
- 21 False
- 23 False
- 25 True
- 27 True
- 29 False

Section 16.2



- 5 150
- 7 54
- 9 $e - 2$
- 11 $3 - \sin 3$
- 13 $(e^4 - 1)(e^2 - 1)e$
- 15 -2.678
- 17 $\int_1^4 \int_1^2 f \, dy \, dx$ or $\int_1^2 \int_1^4 f \, dx \, dy$
- 19 $\int_{-1}^3 \int_{-2}^{(1-3x)/4} f \, dy \, dx$
or $\int_{-2}^1 \int_{-1}^{(1-4y)/3} f \, dx \, dy$